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Micro-welding of non-optical contact sapphire/metal using Burst femtosecond laser

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Highlights

- Micro-welding of non-optical contact sapphire/metal using Burst femtosecond laser was proposed.
- The effective welding of Fe-36Ni with roughness as high as 1.1 μm to sapphire was achieved.
- A welding strength as high as 11 MPa was obtained.

- The welding mechanism for the Burst femtosecond laser is fully revealed.

Abstract

Recently, welding of dissimilar material, such as transparent material to metal material, is widely used in aerospace, microelectronic packaging, medicine, *etc*. Ultrafast laser welding provides a new method for high-performance dissimilar material welding, in which the non-optical contact dissimilar material welding has become a new research hot spot. In this paper, we report the effective welding of Fe-36Ni with roughness as high as 1.1 µm to sapphire, which, to our knowledge, is the roughest dissimilar material welding achieved in ultrafast laser. A 238 fs, 1035 nm, 200 kHz Burst mode femtosecond laser (4 sub-pulses with a total energy of 132.5µJ) yields a smooth and uniform interface with a welding strength up to 11 MPa. In this context, the welding performance and mechanism of Burst mode and single pulse mode ultrafast laser are systematically analyzed. The relevant research results are expected to provide theoretical guidance for the welding of high-strength and high-stability dissimilar materials.

Previous Next

Keywords

Femtosecond laser; Laser welding; Optical contact; Burst mode

1. Introductions

The welding of metal and transparent materials holds significant promise across various industries, including aerospace, electronics, and medical devices [1], [2], [3]. The ability to join these disparate materials allows for the development of lightweight, compact, and high-performance products. In this regard, ultrafast laser welding has seen substantial progress in joining multiple types of materials. Compared with continuous wave laser or long pulsed laser, it provides fine control, efficient energy utilization, and minimal impact on the microstructure of the material, showing significant advantages [4], [5], [6].

Recently, new progress has been made in ultrafast laser welding of dissimilar materials in optical contact. The welding strength has reached 108 MPa [7], and the related welding mechanism has been revealed [8]. However, the welding of dissimilar materials with non-optical contact is still a challenge. The metal plasma excited when ultrafast laser acts on non-optical contact dissimilar material is difficult to be effectively suppressed, and the free expansion of plasma along the interfacial gap makes it difficult to melt and weld the non-optical contact metal and transparent materials [8]. High-rate ultrafast laser has been shown to effectively melt metal materials through the heat accumulation effect, which provides us with new ideas [9].

Therefore, by using a Burst femtosecond laser, this paper carries out the welding of metal materials with roughness higher than 1.1 μm to transparent materials, realizes welding strength up to 11 MPa. In addition, we comprehensively reveal the welding performance and mechanism, and hope to provide new methods for non-optical contact dissimilar material welding.

2. Materials and methods

A 238 fs, 1035 nm, 200 kHz femtosecond laser (Huaray, China) combined with a two-dimensional galvanometer was used to perform the welding experiments. A polished sapphire (melting point 2326 K [10]) of size 10×8×2 mm^3 was placed above Fe-36Ni metal (melting point 1713 K [10]) of size 10×20×2 mm^3 and clamped by a jig (Fig. 1**a**). The rough metal surface morphology is shown in Fig. 1**b**, with grooves of varying depths on the sample surface. The sample roughness (Sa) was as high as 1.16 μm. Using a focusing lens with a focusing length of 80 mm, the focused beam diameter was calculated to be 33 μm. The welding area was set to 3×3 mm^2 and its welding path with a distance between the two welding lines of 80 μm was labeled as shown in Fig. 1**c**. Two modes, single pulse mode with a pulse energy of 132.5 μJ and Burst mode with a total energy of 132.5 μJ (4 sub-pulses), were used to carry out the corresponding process experiments (Fig. 1**d**). The shear strength of the welded sample was measured by a universal testing machine (Fig. 1**e**), and one of the test results is shown in Fig. 1**f**.

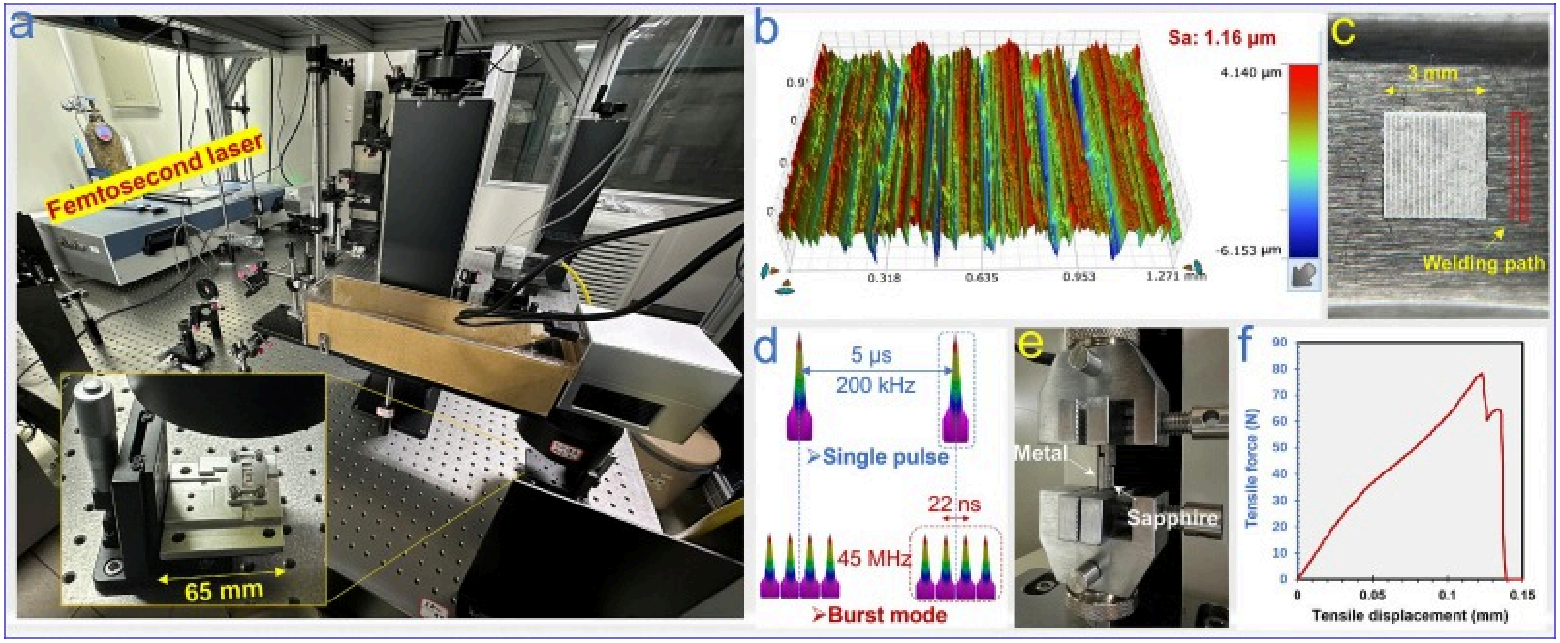

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Fig. 1. **Femtosecond laser welding setup.** (a) Welding system. (b) Surface morphology and roughness of metal. (c) Welded sample. (d) Single pulse mode and Burst mode femtosecond laser. Welding strength (e) testing process and (f) result.

3. Results and discussion

Fig. 2 presents cross-section profiles and surface morphology of the welded samples. The single pulse mode shows a more irregular and disrupted interface (Fig. 2**a**), while the burst mode demonstrates a smoother and more uniform interface (Fig. 2**b**). In addition, the sample welded in single pulse mode shows more cracks in the transparent material since the cracks leads to a reduced transparency. The Burst mode laser welding reduces the drastic fluctuations in the melt pool by releasing energy multiple times in a shorter period of time, reducing the transient effect of each pulse on the material. This mode helps to achieve a more stable melting and solidification process, reducing the generation of microcracks, which in turn improves the integrity of the weld interface. The EDS results of the sample welded by the Burst mode laser indicate distinct layering of elements, with Al and O concentrated at the top (representing the sapphire), and Fe and Ni below, corresponding to the underlying metal. The clearer separation and distribution of elements in the burst mode suggest improved welding quality and better material integration compared to the single pulse mode.

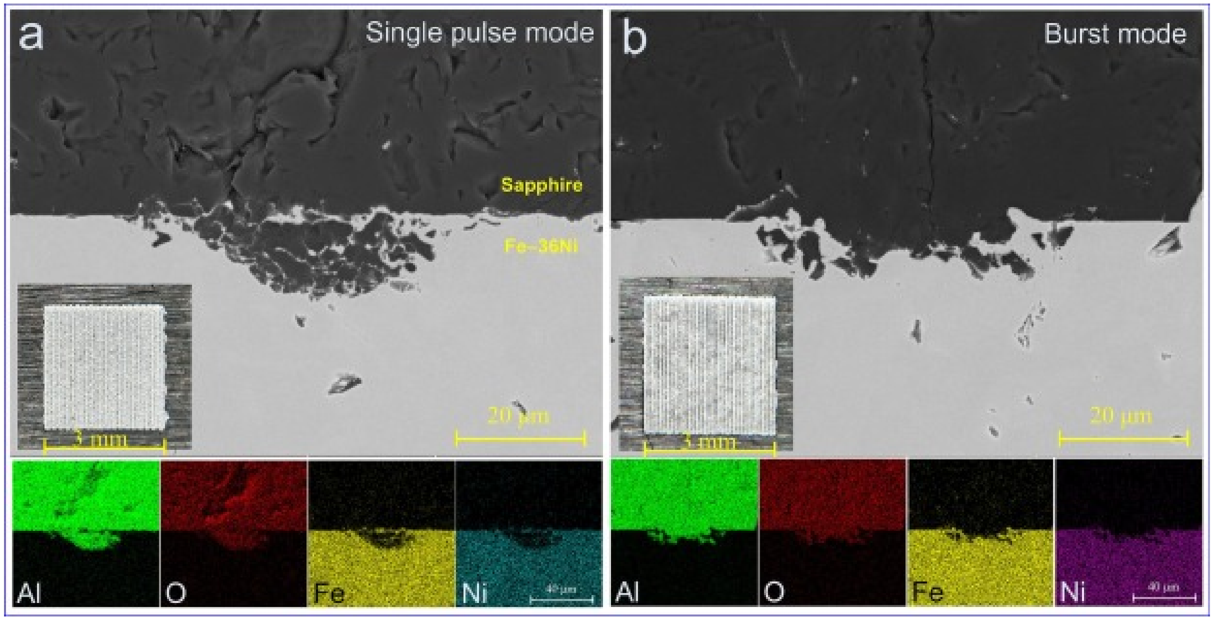

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Fig. 2. **Morphology of welded sample.** Sample welded by the (a) single pulse mode, and (b) Burst mode femtosecond laser, including the EDS results of the welded sample.

With the same total injected energy, the welding strength of the single pulse mode femtosecond laser welding was measured at 11.11 MPa, and the welding strength of the Burst mode femtosecond laser welding was measured at 11.08 MPa. There is significant variability in the profile structure of the welded samples (Fig. 2**a and 2b**), yet the welding strengths are closer, suggesting that the fracture of the samples during the welding strength tests most likely do not occur at the interface of the joint. Fig. 3 shows the fracture surface morphologies of the welded samples under the two welding modes. The morphology of both fracture surfaces, sapphire and Fe-36Ni, suggests that the joint fracture occurred inside the sapphire. The fracture surface shows a relatively uniform texture across the welding area, with a clear boundary at the edge of the welded zone. The overall surface appears to have less pronounced features, which might suggest a more uniform energy distribution during welding (Fig. 2**a, 2a1,**

2b and 2b). Considering the profile morphology of the welded joints in both modes, it can be confirmed that the thermal stresses inside the sapphire after welding are the main cause of fracture. The same total laser energy injection in both modes makes the final thermal stresses generated close to each other. As a result, uniform fracture surfaces appear inside the sapphire.

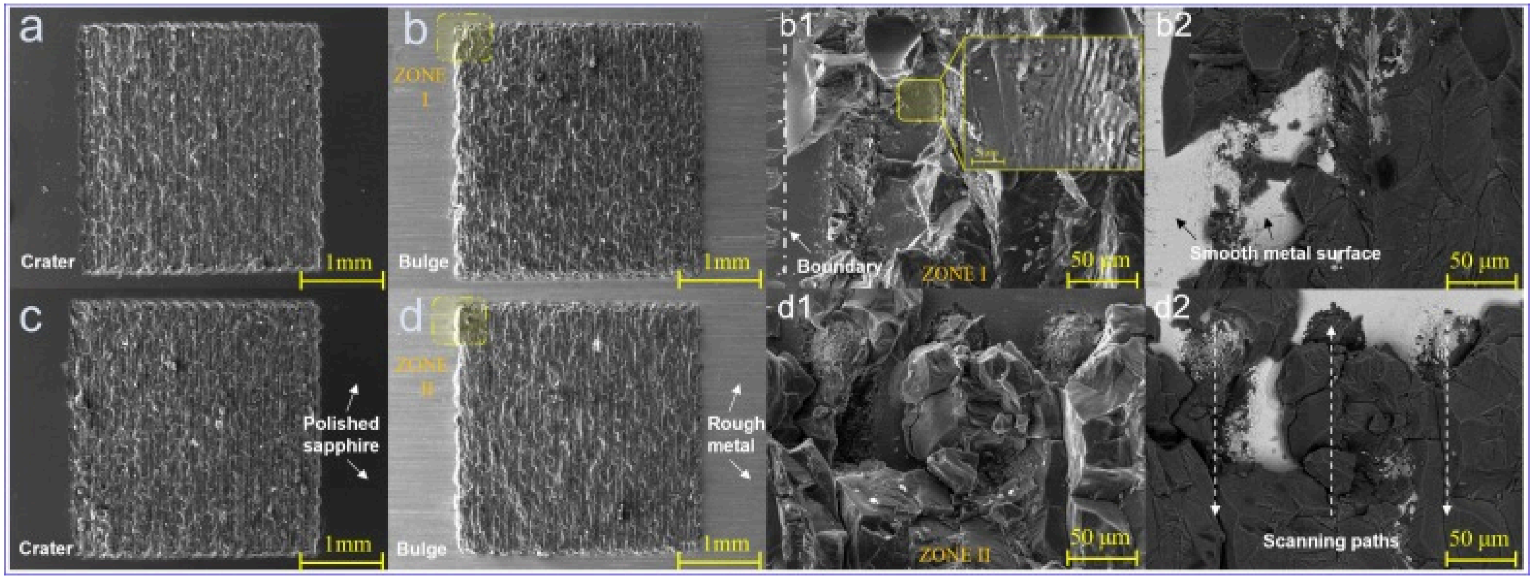

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Fig. 3. **Fracture surfaces.** The fracture surface morphology at sapphire side using the (a)(b) single pulse mode and (c) (d) Burst mode. (a1) (b1). Enlarged SEM images in (b1, b2, d1, and d2) Fe-36Ni side.

Additionally, a partially bare metal surface can be found in the corner of the fracture surface on the metal side (Fig. 2**b and 2b1**). As shown by the labeled boundaries in Fig. 2**b1 and 2b2**, the initial metal surface has melted to form a smooth surface accompanied by localized LIPSS (Laser-induced periodic surface structures) even though the initial metal sample has rough scratches. This suggests that the metal material embodies the typical heating-melting and heating-expansion process that fills up the initial sample gap during the femtosecond laser welding. The path of the laser scan and the morphology of the sapphire remaining on the metal surface show that the highest thermal stresses are formed in the center of the sapphire interior when the femtosecond laser is irradiating on the sapphire.

In terms of the peak power and duration of the femtosecond laser irradiating on the metal target in the different modes, the single pulse mode has a higher peak power density, and thus can penetrate deeper into the metal material, resulting in a greater depth of heating [11], [12]. Also, the interval between pulses is less than the thermal diffusion time of the material, resulting in an effective thermal accumulation. The metal material forms an effective thermal expansion and melting, thus filling the sample gap. At the same time, the shock wave excited by the femtosecond laser in single pulse mode makes part of the melt metal material mobile, resulting in the welded joint shown in Fig. 2**a**. As a result, part of the melt metal flows into the melt sapphire to form an efficient interlocking between the two types of materials (Fig. 4**a**). On the other hand, for the burst mode laser with the same total energy, the reduced peak value results in a lower depth of penetration, however, the duration between the four sub-pulses is as high as 66ns (Fig. 1**d**), and the diameter of the melt metal heated by the heat conduction effect is larger. At this point, the femtosecond laser does not form an interlocking structure at the welding interface (Fig. 4**b**). In addition, the thermal stresses that form inside the transparent material after the ultrafast laser welding are the main cause of fracture and also the main mechanism that limits the welding strength. In this context, the welding performance can be further optimized by reducing the amount of laser energy absorbed by the transparent material (e.g. by reducing the focused spot size).

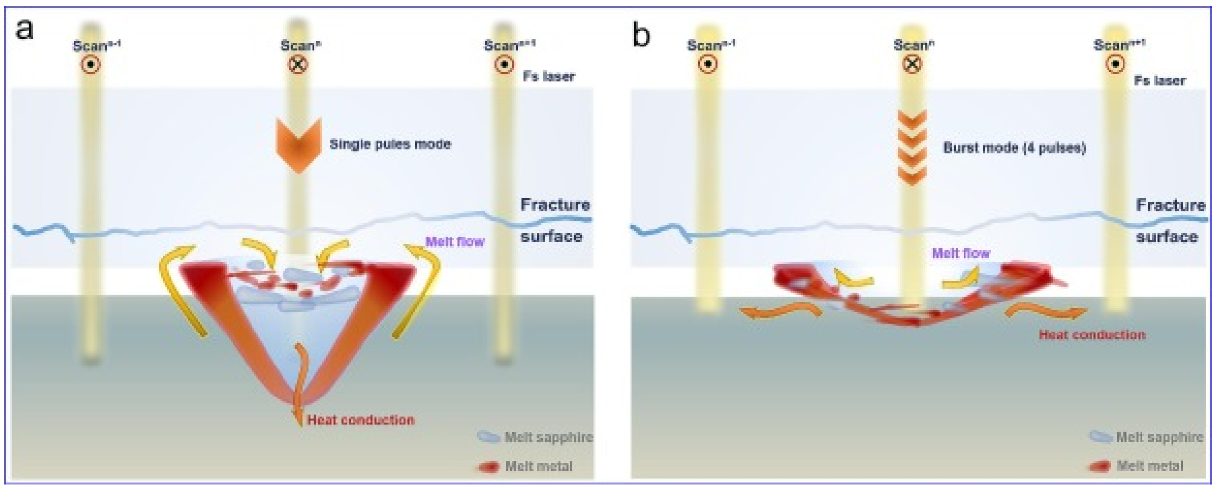

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Fig. 4. **Schematic diagram of the welding mechanism.** (a) Single pulse mode. (b) Burst mode.

4. Conclusion

We explored the welding properties formed by two modes of ultrafast laser, single pulse mode and Burst mode. The single pulse mode forms typical interlocking structure in the welding interface, while the Burst mode forms wider and more stable welded joint. However, the thermal stress inside the transparent material makes the fracture surface after welding in both modes appear on the sapphire side, and the welding strengths obtained in both modes are limited to 11 MPa. In addition, the heating and melting of the metal material by the ultrafast laser is the main mechanism for filling the non-optical contact dissimilar interfaces, which is expected to provide a new theoretical guidance for the welding of transparent/metallic dissimilar materials.

CRediT authorship contribution statement

Zhou Li: Writing – review & editing, Writing – original draft, Supervision, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Zhaoxi Yi:** Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Resources, Methodology, Investigation, Data curation. **Xianshi Jia:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis. **Kai Li:** Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation. **Cong Wang:** Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation. **Ji'an Duan:** Project administration, Methodology, Investigation, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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